A Study of three generalized fixed point theorem

A MINOR PROJECT

submitted to

Department of Mathematics



in partial fulfillment of the requirements for the award of the degree of

Master of Science

in

Mathematics with Computer Science

by

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under the supervision of

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2026

Declaration

I, Mushir Uz Zaman, hereby declare that this MSc project titled **"The Study of three generalized fixed point theorem"** is my original work, carried out under the supervision of Dr. Izhar Uddin, Department of Mathematics, Jamia Millia Islamia.

To the best of my knowledge, this work has not been submitted elsewhere for any degree or diploma. All sources used have been duly acknowledged.

Place: New Delhi

Date:

Mushir Uz Zaman

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Acknowledgement

I express my heartfelt gratitude to my supervisor, Dr. Izhar Uddin, Department of Mathematics, Jamia Millia Islamia, for his constant guidance, encouragement, and valuable feedback throughout this project. His expertise and emphasis on clarity helped me immensely in completing this work.

I am also thankful to Mohammad Saif, a scholar under Dr. Izhar Uddin, for his kind assistance in reviewing and providing feedback on this project. His inputs were valuable in improving my work.

Additionally, I am grateful to my teachers and peers at Jamia Millia Islamia for their support and insightful discussions.

Finally, I thank my family for their unwavering encouragement and understanding during this journey.

Any errors or shortcomings in this project are my own, and I welcome constructive feedback.

Mushir Uz Zaman

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Certificate

On the basis of the declaration submitted by Mushir Uz Zaman (24MCS1036), student of M.Sc. (Mathematics with Computer Science), session 2024–26, I hereby certify that the minor project report titled

"The Study of three generalized fixed point theorem"

which is submitted to the Department of Mathematics, Faculty of Natural Sciences, Jamia Millia Islamia, New Delhi, in partial fulfillment of the requirements for the award of the degree of M.Sc. (Mathematics with Computer Science) has been carried out under my guidance and supervision.

Place: New Delhi **Dr. Izhar Uddin**

Date: (Supervisor)

Preface

This project presents the mathematical investigation into fixed point theory within the framework of complete metric spaces, with a particular focus on the Banach Contraction Principle (BCP) and its generalizations.

The primary objective of this work is to explore, analyze, and understand some significant generalizations of the Banach Contraction Theorem—namely, those proposed by Mier–Keeler, Ciric's generalized contraction, Reich, and Hardy–Rogers—and to investigate their theoretical foundations, differences from the classical result, and illustrative applications.

The project is structured into two main chapters. The first chapter begins by introducing the fundamental concepts of metric spaces, such as completeness, continuity, contraction mappings, and fixed points. This is followed by a rigorous statement and proof of the Banach Fixed Point Theorem. The chapter concludes with detailed applications of the theorem in solving initial value problems, systems of linear algebraic equations, and functional equations.

The second chapter is devoted entirely to generalizations of the Banach Contraction Principle. While studying BCP, we observed that its strict contraction condition may not hold in many real-world problems. This motivated the exploration of more relaxed forms of contraction conditions.

In this chapter, I have presented four landmark generalizations that have shaped modern fixed point theory, namely:

- Theorem of Meir Keeler (1969)
- Ciric generalised contraction
- The Reich Theorem (1973)

-

These generalizations demonstrate the richness and flexibility of fixed point theory and its importance in modern mathematical analysis.

Contents

Declaration **i**

Acknowledgement **ii**

Certificate **iii**

Preface **iv**

1 Banach Fixed Point Theorem and Its Applications **1**

1.1 Preliminaries 1

1.2 The Banach Fixed Point Theorem or Banach Contraction Principle 7

1.3 Applications of the Banach's Theorem to Linear Equations 13

2 Some Generalizations of the Banach Contraction Theorem **21**

2.1 Meir–Keeler Theorem 22

2.2 Ćirić's Generalized Contraction 26

2.3 The Reich and Hardy–Rogers Theorems 33

2.4 Hardy–Rogers Fixed Point Theorem 35

Conclusion **40**

Bibliography **41**

Chapter 1

Banach Fixed Point Theorem and Its Applications

In this chapter, we begin our study by introducing the fundamental concepts and tools necessary for understanding fixed point theory in metric spaces.

We then present the Banach Contraction Principle (also known as the Banach Fixed Point Theorem), a foundational result in nonlinear analysis.

The theorem is stated and proved rigorously, along with several supporting definitions and lemmas.

This chapter lays the theoretical groundwork for the more general fixed point results discussed in the next chapter.

1.1 Preliminaries

Definition 1.1.1. *A metric space is a pair (X, d) where X is a non-empty set and $d : X \times X \to \mathbb{R}$ is a real-valued function (called a metric or distance function) satisfying for all $x, y, z \in X$:*

1. *$d(x, y) \geq 0$ and $d(x, y) = 0$ if and only if $x = y$ (positive definiteness),*

2. $d(x,y) = d(y,x)$ *(symmetry),*

3. $d(x,z) \leq d(x,y) + d(y,z)$ *(triangle inequality).*

Example 1.1.1. *The real numbers* $\mathbb{R}$ *with*

$$d(x,y) = |x-y|$$

form a metric space.

Definition 1.1.2. *Let* X *be a non-empty set and* $d : X \times X \to \mathbb{R}$ *be a real-valued function.*

Then d *is called a pseudo-metric or semi-metric for* X *if the following conditions are satisfied:*

1. $d(x,y) \geq 0, \quad \forall x, y \in X,$

2. $x = y \Rightarrow d(x,y) = 0,$

3. $d(x,y) = d(y,x), \quad \forall x, y \in X$ *(symmetry),*

4. $d(x,z) \leq d(x,y) + d(y,z), \quad \forall x, y, z \in X$ *(triangle inequality).*

Note 1.1.1. *For a pseudo-metric, we may have that* $d(x,y) = 0$ *even if* $x \neq y$.

That is, for a pseudo-metric,

$$x = y \Rightarrow d(x,y) = 0$$

but not conversely.

So, it clearly follows that every metric is a pseudo-metric, but a pseudo-metric is not necessarily a metric.

Definition 1.1.3. *Let* (X,d) *be a metric space and let* $\{x_n\}$ *be a sequence of points in* X.

1. *We say that* $\{x_n\}$ *is a Cauchy sequence if for every* $\epsilon > 0$*, there exists an* $N \in \mathbb{N}$ *such that* $m, n \geq N$ *gives*

$$d(x_m, x_n) < \epsilon.$$

2. *We say that* $\{x_n\}$ *converges to a point* $x \in X$ *if*

$$\lim_{n\to\infty} d(x_n, x) = 0.$$

Definition 1.1.4. *A metric space* (X, d) *is said to be complete if every Cauchy sequence in* X *converges to a point* x *in* X*.*

Example 1.1.2. *The space* $\mathbb{Q}$ *of rational numbers with the standard metric is not complete, while* $\mathbb{R}$ *is complete.*

Definition 1.1.5. *A norm on a (real or complex) vector space* X *is a real-valued function on* X *whose value at* $x \in X$ *is denoted by* $\|x\|$ *(read "norm of* x*") and which has the properties:*

(N1) $\|x\| \geq 0$*,*

(N2) $\|x\| = 0 \iff x = 0$*,*

(N3) $\|\alpha x\| = |\alpha| \cdot \|x\|$*,*

(N4) $\|x + y\| \leq \|x\| + \|y\|$ *(Triangle inequality).*

Here x *and* y *are arbitrary vectors in* X*, and* α *is any scalar.*

A norm on X *defines a metric* d *on* X *which is given by*

$$d(x, y) = \|x - y\|$$

for $x, y \in X$ *and is called the metric induced by the norm.*

The normed space just defined is denoted by $(X, \|\cdot\|)$ *or simply by* X*.*

Definition 1.1.6. *A Banach space is a complete normed space (complete in the metric defined by the norm).*

Examples:

1. **Euclidean space $\mathbb{R}^n$:**

   The space $\mathbb{R}^n$ with the norm

   $$\|x\| = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2}$$

   is a normed space, and it is complete, hence a Banach space.

2. **The space $C[a, b]$:**

   The set of all continuous real-valued functions on the interval $[a, b]$.

   $[a, b]$, equipped with the norm

   $$\|f\|_\infty = \sup_{x \in [a,b]} |f(x)|,$$

   is a Banach space.

   This norm measures the maximum absolute value of a function over the interval.

3. **The sequence space l^p:**

   For $1 \leq p < \infty$, the space of sequences $x = (x_1, x_2, \ldots)$ such that

   $$\|x\|_p = \left(\sum_{i=1}^{\infty} |x_i|^p \right)^{1/p} < \infty$$

   is a Banach space.

4. **The space $L^p([a, b])$:**

   The set of measurable functions $f : [a, b] \to \mathbb{R}$ for which

   $$\|f\|_p = \left(\int_a^b |f(x)|^p dx \right)^{1/p} < \infty$$

   forms a Banach space under pointwise addition and scalar multiplication.

Definition 1.1.7. *Suppose we have a non-empty set X and we have a mapping*

$$T : X \to X$$

(need not be continuous).

Then we say $x \in X$ is a fixed point of T if

$$Tx = x.$$

Definition 1.1.8 (Lipschitz Continuous Function)**.** *Let (X, d) be a metric space.*

A mapping

$$T : X \to X$$

is said to be Lipschitz continuous if there exists a constant $k \geq 0$ such that

$$d(Tx, Ty) \leq kd(x, y)$$

for all $x, y \in X$.

1. *If $k = 1$, then T is said to be non-expansive.*
2. *If $k \in [0, 1)$, then T is said to be contraction.*
3. *If*

$$d(T(x), T(y)) < d(x, y), \quad \forall x \neq y,$$

then T is said to be contractive.

The number k is called the Lipschitz constant of T.

Problem 1.1.1. *Let T be a mapping on $\mathbb{R}$ into itself defined by*

$$T(x) = \frac{1}{2}x.$$

Show that T has a unique fixed point.

Solution:

Given

$$T(x) = \frac{1}{2}x.$$

For a fixed point,

$$T(y) = y.$$

Therefore,

$$\frac{1}{2}y = y.$$

$$\Rightarrow y = 0.$$

Thus, the mapping has a fixed point at $y = 0$.

Now,

$$\|T(x) - T(y)\| = \left\|\frac{1}{2}x - \frac{1}{2}y\right\| = \frac{1}{2}\|x - y\|.$$

Hence, T is a contraction mapping with contraction constant

$$k = \frac{1}{2} < 1.$$

Therefore, by the Banach Fixed Point Theorem, T has a unique fixed point.

Theorem 1.1.1. *Every contraction mapping is continuous.*

Proof. Let $T : X \to X$ be a contraction on a metric space (X, d) with modulus β, and let $x^* \in X$.

Let $\epsilon > 0$ and take

$$\delta = \epsilon.$$

Then for any $x \in X$ satisfying $d(x, x^*) < \delta$, we have:

$$d(Tx, Tx^*) \leq \beta d(x, x^*) < \beta\delta \leq \epsilon.$$

Therefore, T is continuous at x^*.

Since x^* was arbitrary, T is continuous on X.

Theorem 1.1.2. *Every contraction mapping is uniformly continuous.*

Proof. Let $T : X \to X$ be a contraction mapping with modulus β on the metric space (X, d).

Given $\epsilon > 0$, take $\delta = \epsilon$. Then for any $x, y \in X$ with $d(x, y) < \delta$, we have:

$$d(Tx, Ty) \leq \beta\, d(x, y) < \beta\, \delta \leq \epsilon.$$

This inequality holds uniformly for all $x, y \in X$, so T is uniformly continuous.

Theorem 1.1.3 (**Lagrange's Mean Value Theorem**). *Let $f : [a, b] \to \mathbb{R}$ be a function satisfying:*

1. ***Continuity on** $[a, b]$**:** f is continuous on the closed interval [a,b].*
2. ***Differentiability on (a,b):** f is differentiable on the open interval (a,b).*

Then, there exists a point $c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

1.2 The Banach Fixed Point Theorem or Banach Contraction Principle

Theorem 1.2.1 (**Banach fixed point theorem**). *Let (X, d) be a complete metric space and let $T : X \to X$ be a contraction mapping. Then T has a unique fixed point.*

Proof. Let X be a complete metric space and $T : X \to X$ be a contraction mapping.

Let $x_0 \in X$ be any arbitrary point.

If $Tx_0 = x_0$, then there is nothing to prove.

Let $x_0 \neq Tx_0$.

Consider,

$$x_0, x_1, \ldots, x_n, \ldots \in X.$$

$$T(x_0), T(x_1), \ldots, T(x_n), \ldots \in X.$$

Let's define

$$\begin{aligned}
x_1 &= T(x_0), \\
x_2 &= T(x_1) = T^2(x_0), \\
x_3 &= T(x_2) = T^3(x_0), \\
x_4 &= T(x_3) = T^4(x_0), \\
&\vdots \quad \vdots \quad \vdots \\
x_{n+1} &= T(x_n) = T^{n+1}(x_0), \qquad n \geq 0
\end{aligned}$$

Since T is a contraction, therefore, for some $K \in (0, 1)$

$$\begin{aligned}
d(x_{i+1}, x_i) &= d(Tx_i, Tx_{i-1}) \leq Kd(x_i, x_{i-1}) \\
&= Kd(Tx_{i-1}, Tx_{i-2}) \leq K^2 d(x_{i-1}, x_{i-2}) \\
&= K^2 d(Tx_{i-2}, Tx_{i-3}) \leq K^3 d(x_{i-2}, x_{i-3}) \\
&\quad \vdots \qquad\qquad\qquad \vdots \\
&= K^p d(Tx_{i-p}, Tx_{i-p-1}) \leq K^{p+1} d(x_{i-p}, x_{i-p-1}) \\
&\quad \vdots \qquad\qquad\qquad \vdots \\
&= K^{i-1} d(Tx_1, Tx_0) \leq K^i d(x_1, x_0) \\
\implies d(x_{i+1}, x_i) &\leq K^i d(x_1, x_0) \qquad (1.2.1)
\end{aligned}$$

Since X is metric space, therefore, by triangle inequality we have for $m, n \in \mathbb{N}$ and, without

loss of generality, let $m > n$

$$\begin{aligned} d(x_m, x_n) &\leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \cdots + d(x_{n+1}, x_n) \\ &\leq (K^{m-1} + K^{m-2} + \cdots + K^n)\, d(x_1, x_0) \qquad \text{(by using equation (1.2.1))} \\ &= K^n . \frac{1 - K^{m-n}}{1 - K} . d(x_1, x_0) \qquad (1.2.2) \end{aligned}$$

$\because\ 0 < K < 1 \quad \Rightarrow \quad 1 - K^{m-n} < 1.$

Then, equation (1.2.2) becomes

$$d(x_m, x_n) \leq \frac{K^n}{1 - K}\, d(x_1, x_0),$$

$\because\ \lim_{n\to\infty} K^n = 0$ as $K \in (0, 1)$ and $d(x_1, x_0)$ is fixed real number.

Thus, $d(x_m, x_n) \to 0$ as $n \to \infty$

$\Rightarrow \{x_n\}$ is Cauchy sequence in X.

Since, X is complete, therefore, $\{x_n\}$ is convergent in X.

Say,

$$x_n \to x \in X.$$

$$\therefore T(x) \in X.$$

Since, T is a contraction mapping, from the triangle inequality, we have

$$\begin{aligned} d(x, Tx) &\leq d(x, x_n) + d(x_n, Tx) \\ &= d(x, x_n) + d(Tx_{n-1}, Tx) \\ &\leq d(x, x_n) + Kd(x_{n-1}, x) \to 0 \quad \text{as} \quad n \to \infty \\ &\qquad\qquad (\because x_n \to x \quad \& \quad x_{n-1} \to x \quad \text{as} \quad n \to \infty) \\ &\Rightarrow 0 \leq d(x, Tx) \leq 0 \\ &\Rightarrow d(x, Tx) = 0 \iff \boxed{Tx = x} \qquad \text{(by Squeeze or Sandwich Theorem)} \\ &\Rightarrow x \text{ is a } \textbf{fixed point} \text{ of T.} \end{aligned}$$

Uniqueness: Let $Tx = x$ and $Ty = y$ for $x, y \in X$ and $x \neq y$.

Since, T is a contraction mapping, we have

$$d(x,y) = d(Tx,Ty) \leq Kd(x,y) < d(x,y) \qquad (\text{as } K \in (0,1))$$

a contradiction.

Hence, $x = y$.

$$\boxed{or}$$

Since T is a contraction mapping, we have

$$\begin{aligned} d(x,y) = d(Tx,Ty) &\leq K\, d(x,y) && (\text{for } K \in (0,1)) \\ \Rightarrow d(x,y) &\leq K\, d(x,y) \\ \Rightarrow (1-K)\, d(x,y) &\leq 0 \\ \Rightarrow d(x,y) &\leq 0 && (\text{as } 1-K > 0) \\ \Rightarrow 0 \leq d(x,y) &\leq 0 \\ \Rightarrow d(x,y) = 0 &\iff \boxed{x = y}; \end{aligned}$$

a contradiction to our supposition that $x \neq y$.

Hence, x is a unique fixed point of T.

Remark 1.2.1. *If X is not complete in Theorem* (1.2.1) *above, then T may not have a fixed point.*

For example, consider $X = (0,1)$ and the mapping $T : X \to X$ defined by $T(x) = \frac{x}{2}$. Then X is not a complete metric space with the usual metric and T does not have any fixed point. In fact, $T(0) = 0 \notin X$.

Remark 1.2.2. *If T is* ***not contraction*** *in Theorem above, then it may not have a fixed point. For example, consider the metric space $X = [1,\infty)$ with the usual metric and the mapping $T : X \to X$ given by $T(x) = x + \frac{1}{x}$. Then X is a complete metric space but T is not a contraction mapping. In fact,*

$$
\begin{aligned}
|T(x) - T(y)| &= \left| \left(x + \frac{1}{x} \right) - \left(y + \frac{1}{y} \right) \right| \\
&= \left| x + \frac{1}{x} - y - \frac{1}{y} \right| \\
&= |x - y| \left(1 - \frac{1}{xy} \right) \\
&< |x - y|, \qquad \text{for all } x, y \in X.
\end{aligned}
$$

So, T is contractive. Of course, T does not have any fixed point.

Example 1.2.1. *Give an example to show that T satisfies $\|T(x) - T(y)\| = \|x - y\|$ may not have any fixed point?*

Solution:

Let $T : \mathbb{R} \to \mathbb{R}$ be defined by

$$
T(x) = \begin{cases} x - \frac{1}{2}e^x & \text{for } x \leq 0 \\ -\frac{1}{2} + \frac{1}{2}x & \text{for } x \geq 0 \end{cases} \tag{1.2.3}
$$

Now for $x, y \leq 0$

$$
\begin{aligned}
\|T(x) - T(y)\| &= \left\| x - \frac{1}{2}e^x - y + \frac{1}{2}e^y \right\| \\
&= \left\| (x - y) - \frac{1}{2}(e^x - e^y) \right\| \\
&\leq \|x - y\|.
\end{aligned}
$$

For $x, y \geq 0$

$$
\begin{aligned}
\|T(x) - T(y)\| &= \left\| -\frac{1}{2} + \frac{1}{2}x + \frac{1}{2} - \frac{1}{2}y \right\| \\
&= \left\| \frac{1}{2}(x - y) \right\| \\
&\leq \|x - y\|.
\end{aligned}
$$

Thus T satisfies, $\|T(x) - T(y)\| \leq \|x - y\|$. From the definition of fixed point we have $T(x) = x$. Now for $x \leq 0$

$$
T(x) = x
$$

$$\Rightarrow x - \frac{1}{2}e^x = x$$

$$or - \frac{1}{2}e^x = 0$$

$$\Rightarrow e^x = 0,$$

which is not possible for finite x. For $x \geq 0$

$$T(x) = x$$

$$\Rightarrow -\frac{1}{2} + \frac{1}{2}x = x$$

$$\text{or} \quad \frac{1}{2}x = \frac{1}{2}$$

$$\Rightarrow x = -1,$$

is not acceptable as $x \geq 0$. Thus, T defined in (1.2.3) is an example which satisfies the given condition (Banach contraction theorem) but has no fixed point.

The following example shows that if X is a complete metric space and $T : X \to X$ is not a contraction mapping but $T^2 = T \circ T$ is contraction, even then T has a fixed point.

Theorem 1.2.2. *Let (X, d) be a complete metric space and $T : X \to X$ be a mapping such that for some integer m, $T^m = \underbrace{T \circ T \circ \cdots \circ T}_{m \text{ times}}$ is a contraction mapping. Then T has a unique fixed point.*

Proof. By above theorem, T^m has a unique fixed point $x \in X$ that is, $T^m(x) = x$.

Then

$$T(x) = T(T^m(x)) = T^m(T(x))$$

and so $T(x)$ is a fixed point of T^m. Since the fixed point of T^m is unique, so $T(x) = x$.

Uniqueness: we assume that y is another fixed point of T.

Then $T(y) = y$ and so $T^m(y) = y$.

Again by the uniqueness of the fixed point of T^m, we have $x = y$.

Hence, $x \in X$ is a unique fixed point of T.

Remark 1.2.3. *The continuity condition on T is not necessary.*

Example 1.2.2. *Let $X = \mathbb{R}$ be a metric space with the usual metric and $T : X \to X$ be*

a mapping defined as

$$T(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \in \mathbb{Q}^c. \end{cases}$$

Then, T is not continuous and, hence, not a contraction mapping. Now

$$T^2(x) = T(T(x)) = \begin{cases} T(1) = 1 & \text{if } x \in \mathbb{Q} \\ T(0) = 1 & \text{if } x \in \mathbb{Q}^c. \end{cases}$$

Then, T^2 is a contraction mapping, but both T^2 and T have the same fixed point that is 1.

1.3 Applications of the Banach's Theorem to Linear Equations

One of the most natural and computationally significant applications of the Banach Contraction Principle lies in solving systems of linear equations of the form

$$Ax = c.$$

Rather than computing the inverse of A directly, we transform the system into a fixed point equation of the form

$$x = Cx + b,$$

and then apply the Banach Contraction Principle iteratively to obtain the unique solution.

Theorem 1.3.1 (**Kreyszig Theorem**). *Consider a system of n linear equations in n unknowns:*

$$x = Cx + b \tag{1.3.1}$$

where $C = (c_{jk})$ is a fixed real $n \times n$ matrix and $b \in \mathbb{R}^n$ is a fixed vector.

If the row sum condition

$$\sum_{k=1}^{n} |c_{jk}| < 1, \quad j = 1, 2, \ldots, n \tag{1.3.2}$$

is satisfied, then the system (1.3.1) has precisely one solution x^*.

This solution can be obtained as the limit of the iterative sequence

$$x^{(0)}, x^{(1)}, x^{(2)}, \dots$$

where $x^{(0)}$ is arbitrary and

$$x^{(m+1)} = Cx^{(m)} + b, \quad m = 0, 1, 2, \dots \tag{1.3.3}$$

Moreover, define

$$\alpha = \max_j \sum_{k=1}^{n} |c_{jk}| < 1.$$

Then the error estimates are given by

$$d(x^{(m)}, x^*) \leq \frac{\alpha}{1-\alpha}\, d(x^{(m-1)}, x^{(m)}) \leq \frac{\alpha^m}{1-\alpha}\, d(x^{(0)}, x^{(1)}). \tag{1.3.4}$$

Application 1.3.1. *Find the solution of the system of n linear algebraic equations with n unknowns:*

$$\left.\begin{aligned} &a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ &a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ &\vdots \\ &a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n \end{aligned}\right\} \tag{1.3.4}$$

Solution:

The system (1.3.4) can be written as

$$\left.\begin{aligned} x_1 &= (1 - a_{11})x_1 - a_{12}x_2 - \dots - a_{1n}x_n + b_1 \\ x_2 &= -a_{21}x_1 + (1 - a_{22})x_2 - \dots - a_{2n}x_n + b_2 \\ &\vdots \\ x_n &= -a_{n1}x_1 - a_{n2}x_2 - \dots + (1 - a_{nn})x_n + b_n \end{aligned}\right\} \tag{1.3.5}$$

Let $a_{ij} = -a_{ij} + \delta_{ij}$, where

$$\delta_{ij} = \begin{cases} 1 & \text{for } i = j \\ 0 & \text{for } i \neq j. \end{cases}$$

Then the equation (1.3.5) can be written in the following equivalent form:

$$x_i = \sum_{j=1}^{n} a_{ij} x_j + b_i, \quad i = 1, 2, \ldots, n \tag{1.3.6}$$

If $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$, then equation (1.3.6) can be written in the form $Tx = x$, where T is defined by

$$Tx = y, \tag{1.3.7}$$

where $y = (y_1, y_2, \ldots, y_n)$ and $y_i = \sum_{j=1}^{n} a_{ij} x_j + b_i$. Here $T : \mathbb{R}^n \to \mathbb{R}^n$ and (a_{ij}) is an $(n \times n)$ matrix. Finding solutions of the system (1.3.4) or (1.3.5) is thus equivalent to finding the fixed point of operator (1.3.7).

To find a unique fixed point of T (that is, a unique solution of equation (1.3.7), we apply the Banach contraction principle. Equation (1.3.7) has a unique solution if

$$\sum_{j=1}^{n} |a_{ij}| = \sum_{j=1}^{n} | - a_{ij} + \delta_{ij}| \leq k < 1, \quad i = 1, 2, \ldots, n$$

For

$$x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n, \quad x' = (x'_1, x'_2, \ldots, x'_n) \in \mathbb{R}^n$$

$$y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n, \quad y' = (y'_1, y'_2, \ldots, y'_n) \in \mathbb{R}^n$$

We have

$$\|Tx - Tx'\| = \|y - y'\|$$

where

$$y'_i = \sum_{j=1}^{n} a_{ij} x'_j + b_i, \quad i = 1, 2, \ldots, n$$

Also, if $y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$, then $\|y\| = \sup_{1 \leq i \leq n} |y_i|$.

Therefore,

$$\begin{aligned}\|Tx - Tx'\| &= \|y - y'\| \\ &= \sup_{1\le i\le n} |y_i - y_i'| \\ &= \sup_{1\le i\le n} \left| \sum_{j=1}^{n} a_{ij}x_j + b_i - \sum_{j=1}^{n} a_{ij}x_j' - b_i \right| \\ &= \sup_{1\le i\le n} \left| \sum_{j=1}^{n} a_{ij}(x_j - x_j') \right|\end{aligned}$$

Using the triangle inequality we get

$$\begin{aligned}\|Tx - Tx'\| &\le \sup_{1\le i\le n} \sum_{j=1}^{n} |a_{ij}||x_j - x_j'| \\ &\le \sup_{1\le i\le n} \sum_{j=1}^{n} |a_{ij}| \cdot \sup_{1\le j\le n} |x_j - x_j'| \\ &\le k\|x - x'\|, \quad \text{where} \sum_{j=1}^{n} |a_{ij}| \le k < 1\end{aligned}$$

where $\|x - x'\| = \sup_{1\le j\le n} |x_j - x_j'|$.

This shows that T is a contraction mapping of the Banach space into itself.

Hence, by the Banach Contraction Principle, there exists a unique fixed point x^* of T in $\mathbb{R}^n$, that is, x^* is a solution of (1.3.4).

Application 1.3.2 (**Application of Banach's Theorem to Differential Equations**). *The most interesting applications of Banach's Fixed Point Theorem arise in connection with function spaces. The theorem yields existence and uniqueness results for differential equations and provides a powerful method for obtaining solutions through successive approximations.*

Let us consider the initial value problem

$$x'(t) = \frac{dx}{dt} = f(t, x), \qquad x(t_0) = x_0. \tag{1.4.1}$$

where $f : A \subset \mathbb{R}^2 \to \mathbb{R}$ is a given function and $x(t)$ is an unknown function which we wish to determine.

In this subsection, we use Banach's Fixed Point Theorem to prove the famous Picard–

Lindelöf Theorem, which guarantees the existence and uniqueness of a solution to the initial value problem.

Theorem 1.3.2 (Picard–Lindelöf)**.** *Let f be continuous on a rectangle*

$$R = \{(t, x) : |t - t_0| \leq a,\ |x - x_0| \leq b\},$$

and therefore bounded on R, say

$$|f(t, x)| \leq c.$$

Suppose that f satisfies a Lipschitz condition on R with respect to its second argument, meaning that there exists a constant k such that

$$|f(t, x) - f(t, y)| \leq k|x - y|$$

for all $(t, x), (t, y) \in R$.

Then the initial value problem (1.4.1) has a unique solution on an interval

$$[t_0 - \beta, t_0 + \beta],$$

where

$$\beta < \min\left\{a,\ \frac{b}{c},\ \frac{1}{k}\right\}. \tag{1.4.2}$$

Proof. We split the proof into five steps. **Step 1: Equivalent formulation as an integral equation**

Suppose that

$$x \in C^1[t_0 - a, t_0 + a]$$

is a solution of (1.4.1). Integrating both sides from t_0 to t, we obtain

$$x(t) = x_0 + \int_{t_0}^{t} f(s, x(s))\, ds. \tag{1.4.3}$$

Conversely, if

$$x \in C[t_0 - a, t_0 + a]$$

satisfies (1.4.3), then by the Fundamental Theorem of Calculus, x is continuously differentiable and satisfies (1.4.1).

Hence the initial value problem (1.4.1) is equivalent to the integral equation (1.4.3).

Step 2: Constructing an operator

Let

$$J = [t_0 - \beta, t_0 + \beta]$$

and for every $y \in C(J)$ define

$$T(y)(t) = x_0 + \int_{t_0}^{t} f(s, y(s))\, ds, \qquad t \in J.$$

Consider the set

$$X = \left\{ y \in C(J) : y(t_0) = x_0,\ \sup_{t \in J} |x_0 - y(t)| \leq c\beta \right\}.$$

This is a closed subspace of $C(J)$ endowed with the metric

$$d_\infty(x, y) = \sup_{t \in J} |x(t) - y(t)|.$$

Therefore (X, d_∞) is a complete metric space.

Step 3: Showing that $T : X \to X$

Let $y \in X$.

Clearly,

$$T(y)(t_0) = x_0.$$

Also,

$$
\begin{aligned}
|x_0 - T(y)(t)| &= \left| \int_{t_0}^{t} f(s, y(s))\, ds \right| \\
&\leq |t - t_0| \max_{s \in J} |f(s, y(s))| \\
&\leq c\beta.
\end{aligned}
$$

Hence

$$T(y) \in X.$$

Therefore

$$T : X \to X.$$

Step 4: Showing that T is a contraction

Let

$$y_1, y_2 \in X.$$

Then

$$
\begin{aligned}
|T(y_1)(t) - T(y_2)(t)| &= \left| \int_{t_0}^{t} \big(f(s, y_1(s)) - f(s, y_2(s))\big) ds \right| \\
&\leq |t - t_0| \max_{s \in J} |f(s, y_1(s)) - f(s, y_2(s))| \\
&\leq k\beta \sup_{s \in J} |y_1(s) - y_2(s)|.
\end{aligned}
$$

Taking the supremum over $t \in J$, we obtain

$$d_\infty(T(y_1), T(y_2)) \leq k\beta\, d_\infty(y_1, y_2).$$

Since by (1.4.2)

$$k\beta < 1,$$

the operator T is a contraction on X.

Step 5: Conclusion

By Banach's Fixed Point Theorem, the operator T possesses a unique fixed point

$$x \in X$$

such that

$$T(x) = x.$$

Therefore

$$x(t) = x_0 + \int_{t_0}^{t} f(s, x(s))\, ds.$$

By Step 1, this fixed point is precisely the unique solution of the initial value problem (1.4.1) on the interval

$$[t_0 - \beta, t_0 + \beta].$$

Hence the initial value problem has a unique continuous solution.

Banach's Contraction Principle can also be applied to systems of linear algebraic equations. By transforming a system into a fixed point problem, one can guarantee the existence and uniqueness of solutions under suitable matrix conditions. This demonstrates the wide applicability of the contraction framework in analysis and applied mathematics.

Chapter 2

Some Generalizations of the Banach Contraction Theorem

In the previous chapter, we explored the Banach Contraction Principle (BCP) in detail, along with its foundational definitions, important lemmas, and practical applications. While studying these results, it became evident that many mappings encountered in practice do not satisfy the strict contraction condition required in the Banach Contraction Principle. This observation motivated mathematicians to develop various generalizations of BCP that weaken the contraction requirement while still ensuring the existence of fixed points.

In this chapter, we discuss some important generalizations of the Banach Contraction Principle, including the *Meir–Keeler*, *Ćirić*, *Reich*, and *Hardy–Rogers* contractions. These generalizations significantly broaden the scope of fixed point theory and provide a deeper understanding of the behavior of nonlinear mappings.

For example, the Meir–Keeler contraction, introduced in 1969, replaces the standard Lipschitz-type contraction condition with a more flexible inequality, thereby extending the applicability of fixed point results to a wider class of mappings.

Such developments and generalizations can be found in standard references such as Introductory Functional Analysis with Applications by KREYSZIG and Advanced Functional Analysis by EBERHARD MALKOWSKY AND VLADIMIR RAKOCEVIC, who have made significant contributions to the development of fixed point theory.

We now proceed to study these generalizations individually.

2.1 Meir–Keeler Theorem

Meir–Keeler theorem introduced by Meir and Keeler in 1969, was motivated by the desire to extend to a more general class of contractions by relaxing the condition of contraction.

Definition 2.1.1. *Let (X, d) be a metric space. The function*

$$f : X \to X$$

is said to be a weakly uniformly strict contraction, or a Meir–Keeler contraction (MK contraction) if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\varepsilon \leq d(x, y) < \varepsilon + \delta \implies d(f(x), f(y)) < \varepsilon.$$

Theorem 2.1.1 (Meir and Keeler)**.** *Let (X, d) be a complete metric space and*

$$f : X \to X$$

be a function. If the above condition is satisfied, then f has a unique fixed point z. Moreover, for each $x \in X$,

$$\lim_{n\to\infty} f^n(x) = z.$$

Proof. First we note that the MK condition implies that f is a contractive map, that is,

$$x \neq y \implies d(f(x), f(y)) < d(x, y).$$

Hence f is a continuous function and has at most one fixed point.

We note that if $(f^n(x))$ is a Cauchy sequence for each $x \in X$, then the function f has a unique fixed point, and the limit condition is satisfied.

Since X is a complete space, every Cauchy sequence $(f^n(x))$ has a limit $z(x)$.

The continuity of f implies

$$f(z(x)) = f\left(\lim_{n\to\infty} f^n(x)\right) = \lim_{n\to\infty} f^{n+1}(x) = z(x).$$

Hence $z(x)$ is the unique fixed point of f.

The proof will be complete if we show that

$$(f^n(x)) = (x_n)$$

is a Cauchy sequence for each $x \in X$.

Let

$$c_n = d(x_n, x_{n+1}).$$

Then (c_n) is a decreasing sequence and

$$\lim_{n\to\infty} c_n = 0.$$

We assume that there exists a sequence (x_n) which is not a Cauchy sequence.

Then there exists $2\varepsilon > 0$ such that, for each $m_0 \in \mathbb{N}$, there exist $n, m \in \mathbb{N}$ with $n, m > m_0$ and

$$d(x_m, x_n) > 2\varepsilon.$$

It follows that there exists $\delta > 0$ such that

$$\varepsilon \leq d(x, y) < \varepsilon + \delta \implies d(f(x), f(y)) < \varepsilon.$$

The implication in the previous condition remains true if we replace δ by

$$\delta' = \min\{\delta, \varepsilon\}.$$

Let $m_0 \in \mathbb{N}$ be such that

$$c_{m_0} < \frac{\delta'}{3},$$

and let $m, n > m_0$ be such that $m < n$ and

$$d(x_m, x_n) > 2\varepsilon.$$

We prove that there exists

$$j \in \{m, m+1, \ldots, n\}$$

such that

$$\varepsilon + \frac{2\delta'}{3} < d(x_m, x_j) < \varepsilon + \delta'.$$

To prove this, we note that

$$d(x_{n-1}, x_n) < \frac{\delta'}{3}.$$

Since

$$d(x_m, x_n) > 2\varepsilon$$

and

$$d(x_m, x_n) \leq d(x_m, x_{n-1}) + d(x_{n-1}, x_n),$$

it follows that

$$\varepsilon + \frac{2\delta'}{3} < d(x_m, x_{n-1}).$$

Let k be the smallest natural number in

$$\{m, m+1, \ldots, n\}$$

such that

$$\varepsilon + \frac{2\delta'}{3} < d(x_m, x_k)$$

holds.

We prove that

$$d(x_m, x_k) < \varepsilon + \delta'.$$

If not, then

$$\varepsilon + \delta' \leq d(x_m, x_k) \leq d(x_m, x_{k-1}) + d(x_{k-1}, x_k) < d(x_m, x_{k-1}) + \frac{\delta'}{3},$$

that is,

$$\varepsilon + \frac{2\delta'}{3} < d(x_m, x_{k-1}).$$

This is a contradiction to the minimality of k.

Therefore the first inequality holds.

Now,

$$d(x_m, x_k) \leq d(x_m, x_{m+1}) + d(x_{m+1}, x_{k+1}) + d(x_{k+1}, x_k),$$

and the MK condition implies

$$d(x_m, x_k) \leq c_m + \varepsilon + c_k < \frac{\delta'}{3} + \varepsilon + \frac{\delta'}{3}.$$

This is a contradiction.

Hence (x_n) is a Cauchy sequence.

Example 2.1.1. *Let*

$$X = [0, \infty)$$

with the usual metric and define

$$f(x) = x - \ln(1 + x).$$

Then f is a Meir–Keeler contraction but not a Banach contraction.

Verification:

Note that

$$f(0) = 0,$$

so 0 is a fixed point.

For any $x \neq y$, by Mean Value Theorem,

$$d(f(x), f(y)) = |f(x) - f(y)| = \left|\frac{x - y}{1 + c}\right| < |x - y| = d(x, y),$$

for some c between x and y.

So f is contractive.

Why not Banach:

The ratio

$$\frac{d(f(x), f(y))}{d(x, y)} = \frac{1}{1 + c} \to 1 \quad \text{as } x, y \to \infty.$$

Hence no single

$$q \in [0, 1)$$

works globally.

Therefore f is not a Banach contraction.

Why Meir–Keeler:

Given $\varepsilon > 0$, choose

$$\delta = \varepsilon \cdot \frac{\varepsilon}{1+\varepsilon}.$$

Then

$$\varepsilon \leq d(x,y) < \varepsilon + \delta$$

implies

$$d(f(x), f(y)) < \varepsilon.$$

Hence the Meir–Keeler condition is satisfied.

Therefore f has a unique fixed point

$$z = 0.$$

2.2 Ćirić's Generalized Contraction

Ćirić generalized the well-known contractive condition and introduced the following concept.

Definition 2.2.1. *Let (X,d) be a metric space. A mapping $f : X \to X$ is a λ-generalized contraction if, for all $x, y \in X$, there exist nonnegative numbers $q(x,y)$, $r(x,y)$, $s(x,y)$ and $t(x,y)$ such that*

$$\sup_{x,y \in X} \{q(x,y) + r(x,y) + s(x,y) + 2t(x,y)\} = \lambda < 1,$$

and for all $x, y \in X$,

$$\begin{aligned} d(f(x), f(y)) \leq\ & q(x,y)d(x,y) + r(x,y)d(x, f(x)) \\ & + s(x,y)d(y, f(y)) \\ & + t(x,y)\big(d(x, f(y)) + d(y, f(x))\big). \end{aligned}$$

This condition is equivalent to: there exists $h \in (0,1)$ such that for all $x, y \in X$,

$$d(f(x), f(y)) \leq h \max \left\{ d(x,y), d(x, f(x)), d(y, f(y)), \frac{d(x, f(y)) + d(y, f(x))}{2} \right\}.$$

Example 2.2.1. *Let*

$$X = [0, 2] \subseteq \mathbb{R}$$

and define

$$f(x) = \begin{cases} \dfrac{x}{9}, & 0 \leq x \leq 1, \\ \dfrac{x}{10}, & 1 < x \leq 2. \end{cases}$$

Banach contraction fails:

Take

$$x = \frac{999}{1000}, \qquad y = \frac{1001}{1000}.$$

Then

$$f(x) = \frac{111}{1000}, \qquad f(y) = \frac{1001}{10000}.$$

Hence,

$$\begin{aligned} d(f(x), f(y)) &= \left| \frac{111}{1000} - \frac{1001}{10000} \right| \\ &= \frac{109}{10000} = \frac{981}{90000}. \end{aligned}$$

Also,

$$d(x, y) = \left| \frac{999}{1000} - \frac{1001}{1000} \right| = \frac{2}{1000} = \frac{180}{90000}.$$

Therefore,

$$d(f(x), f(y)) = \frac{981}{90000} > 5 \cdot \frac{180}{90000} = 5\, d(x, y).$$

So no fixed $q \in [0, 1)$ works globally. Hence f is not a Banach contraction.

Ćirić condition is satisfied:

The condition holds with

$$q(x,y) = \frac{1}{10}, \qquad r(x,y) = s(x,y) = \frac{1}{4}, \qquad t(x,y) = \frac{1}{6}$$

for all $x, y \in X$.

Now,

$$q(x,y) + r(x,y) + s(x,y) + 2t(x,y) = \frac{1}{10} + \frac{1}{4} + \frac{1}{4} + 2\left(\frac{1}{6}\right).$$

That is,

$$= \frac{1}{10} + \frac{1}{2} + \frac{1}{3} = \frac{3+15+10}{30} = \frac{28}{30} = \frac{14}{15} < 1.$$

Hence all the conditions of Ćirić's generalized contraction are satisfied.

Therefore, f *is a Ćirić generalized contraction, although it is not a Banach contraction.*

Theorem 2.2.1 (**Ćirić**). *If* $f : X \to X$ *is a* λ*-generalized contraction on an* f*-orbitally complete metric space* X*, then, for any* $x \in X$*, the iterative sequence* $(f^n(x))$ *converges to the unique fixed point* z *of* f*, and*

$$d(f^n(x), z) \leq \frac{\lambda^n}{1-\lambda} \cdot d(x, f(x)).$$

Proof. For an arbitrary $x \in X$, we define the sequence (x_n) by

$$x_0 = x$$

and

$$x_n = f(x_{n-1}), \qquad n \in \mathbb{N}.$$

Then we obtain from the definition:

$$\begin{aligned} d(x_n, x_{n+1}) &= d(f(x_{n-1}), f(x_n)) \\ &\leq q(x_{n-1}, x_n)d(x_{n-1}, x_n) \\ &\quad + r(x_{n-1}, x_n)d(x_{n-1}, f(x_{n-1})) \\ &\quad + s(x_{n-1}, x_n)d(x_n, f(x_n)) \\ &\quad + t(x_{n-1}, x_n)\big(d(x_{n-1}, f(x_n)) + d(x_n, f(x_{n-1}))\big). \end{aligned}$$

Since

$$f(x_{n-1}) = x_n$$

and

$$f(x_n) = x_{n+1},$$

we get

$$\begin{aligned} d(x_n, x_{n+1}) &\leq q(x_{n-1}, x_n)d(x_{n-1}, x_n) \\ &\quad + r(x_{n-1}, x_n)d(x_{n-1}, x_n) \\ &\quad + s(x_{n-1}, x_n)d(x_n, x_{n+1}) \\ &\quad + t(x_{n-1}, x_n)d(x_{n-1}, x_{n+1}). \end{aligned}$$

Moreover,

$$d(x_{n-1}, x_{n+1}) \leq d(x_{n-1}, x_n) + d(x_n, x_{n+1}).$$

Hence,

$$\begin{aligned} d(x_n, x_{n+1}) &\leq (q(x_{n-1}, x_n) + r(x_{n-1}, x_n))d(x_{n-1}, x_n) \\ &\quad + s(x_{n-1}, x_n)d(x_n, x_{n+1}) \\ &\quad + t(x_{n-1}, x_n)(d(x_{n-1}, x_n) + d(x_n, x_{n+1})). \end{aligned}$$

Therefore,

$$\begin{aligned} d(x_n, x_{n+1}) &\leq (q(x_{n-1}, x_n) + r(x_{n-1}, x_n) + t(x_{n-1}, x_n))d(x_{n-1}, x_n) \\ &\quad + (s(x_{n-1}, x_n) + t(x_{n-1}, x_n))d(x_n, x_{n+1}). \end{aligned}$$

So we have

$$d(x_n, x_{n+1}) \leq \frac{q(x_{n-1}, x_n) + r(x_{n-1}, x_n) + t(x_{n-1}, x_n)}{1 - s(x_{n-1}, x_n) - t(x_{n-1}, x_n)} d(x_{n-1}, x_n).$$

Because of

$$q(x, y) + r(x, y) + t(x, y) + \lambda s(x, y) + \lambda t(x, y) \leq \lambda,$$

we get

$$\frac{q(x, y) + r(x, y) + t(x, y)}{1 - s(x, y) - t(x, y)} \leq \lambda \qquad \text{for all } x, y \in X.$$

Hence,

$$d(x_n, x_{n+1}) \leq \lambda d(x_{n-1}, x_n).$$

Repeating this process, we obtain

$$d(x_n, x_{n+1}) \leq \lambda^n d(x, f(x)).$$

Now for all $m \geq n$,

$$\begin{aligned} d(x_n, x_m) &\leq \sum_{k=n}^{m-1} d(x_k, x_{k+1}) \\ &\leq \sum_{k=n}^{m-1} \lambda^k d(x, f(x)). \end{aligned}$$

Therefore,

$$d(x_n, x_{n+p}) \leq \frac{\lambda^n}{1 - \lambda} d(x, f(x)).$$

Hence (x_n) is a Cauchy sequence in $O(x)$.

Since X is f-orbitally complete, there exists $z \in X$ such that

$$x_n \to z.$$

It remains to show that

$$f(z) = z.$$

Now,

$$\begin{aligned} d(f(z), f(x_n)) \leq\ & q(z, x_n)d(z, x_n) \\ & + r(z, x_n)\big(d(z, x_{n+1}) + d(x_{n+1}, f(z))\big) \\ & + s(z, x_n)d(x_n, x_{n+1}) \\ & + t(z, x_n)\big(d(z, x_{n+1}) + d(f(z), x_n)\big). \end{aligned}$$

Using the generalized contraction condition again, we get

$$\begin{aligned} d(f(z), f(x_n)) \leq\ & \lambda d(z, x_n) + (r(z, x_n) + t(z, x_n))d(z, x_{n+1}) \\ & + (r(z, x_n) + t(z, x_n))d(f(z), f(x_n)) \\ & + \lambda d(x_n, x_{n+1}). \end{aligned}$$

Hence,

$$\begin{aligned} d(f(z), f(x_n)) \leq\ & \lambda\big(d(z, x_n) + d(z, x_{n+1}) + d(x_n, x_{n+1})\big) \\ & + \lambda d(f(z), f(x_n)). \end{aligned}$$

Therefore,

$$d(f(z), f(x_n)) \leq \frac{\lambda}{1-\lambda}\left[d(z, x_n) + d(z, x_{n+1}) + d(x_n, x_{n+1})\right].$$

Taking limit as $n \to \infty$, we obtain

$$d(f(z), z) = 0.$$

Hence,

$$f(z) = z.$$

Thus z is a fixed point of f.

Uniqueness: Suppose z and w are both fixed points of f, that is,

$$f(z) = z \qquad \text{and} \qquad f(w) = w.$$

Then,

$$\begin{aligned} d(z,w) &= d(f(z), f(w)) \\ &\leq q(z,w)d(z,w) \\ &\quad + r(z,w)d(z, f(z)) \\ &\quad + s(z,w)d(w, f(w)) \\ &\quad + t(z,w)\big(d(z, f(w)) + d(w, f(z))\big). \end{aligned}$$

Since

$$f(z) = z \qquad \text{and} \qquad f(w) = w,$$

we get

$$\begin{aligned} d(z,w) &\leq q(z,w)d(z,w) + t(z,w)(d(z,w) + d(w,z)) \\ &= (q(z,w) + 2t(z,w))d(z,w). \end{aligned}$$

Since

$$q(z,w) + 2t(z,w) \leq \lambda < 1,$$

it follows that

$$d(z,w) \leq \lambda d(z,w).$$

This is possible only if

$$d(z,w) = 0.$$

Hence,

$$z = w.$$

Therefore, the fixed point of f is unique.

2.3 The Reich and Hardy–Rogers Theorems

In 1971, Reich proved the following theorem which generalizes Banach's and Kannan's theorems. We note that for $a = b = 0$, we obtain Banach's theorem, and for $a = b$ and $c = 0$, we obtain Kannan's theorem.

Theorem 2.3.1 (**Reich**). *Let (X, d) be a complete metric space and let*

$$f : X \to X$$

be a map for which there exist nonnegative numbers a, b and c with

$$a + b + c < 1$$

such that for all $x, y \in X$,

$$d(f(x), f(y)) \leq ad(x, f(x)) + bd(y, f(y)) + cd(x, y).$$

Then the map f has a unique fixed point.

Proof. Let $x \in X$. We consider the sequence

$$(f^n(x)).$$

If we put

$$x = f^n(x) \qquad \text{and} \qquad y = f^{n-1}(x)$$

in the above inequality, then we have for all $n \geq 1$,

$$\begin{aligned} d(f(f^n(x)), f(f^{n-1}(x))) \leq\ & ad(f^n(x), f(f^n(x))) \\ & + bd(f^{n-1}(x), f(f^{n-1}(x))) \\ & + cd(f^n(x), f^{n-1}(x)). \end{aligned}$$

Hence, we obtain

$$d(f^{n+1}(x), f^n(x)) \leq pd(f^n(x), f^{n-1}(x)),$$

where

$$0 \leq p = \frac{b+c}{1-a} < 1.$$

It follows that

$$d(f^{n+1}(x), f^n(x)) \leq p^n d(x, f(x)).$$

For every $m > n$,

$$d(f^m(x), f^n(x)) \leq \frac{p^n}{1-p}\, d(x, f(x)).$$

Thus $(f^n(x))$ is a Cauchy sequence, and there exists

$$z \in X$$

with

$$z = \lim_{n\to\infty} f^n(x).$$

We are going to show that $f(z) = z$. It suffices to show that

$$\lim_{n\to\infty} f^{n+1}(x) = f(z).$$

When we choose

$$x = f^n(x) \qquad \text{and} \qquad y = z$$

in the contractive condition, then we have for all $n \geq 1$,

$$\begin{aligned} d(f^{n+1}(x), f(z)) \leq\ & ad(f^n(x), f^{n+1}(x)) + bd(z, f(z)) \\ & + cd(f^n(x), z). \end{aligned}$$

Also,

$$\begin{aligned} d(z, f(z)) &\leq d(z, f^{n+1}(x)) + d(f^{n+1}(x), f(z)) \\ &\leq d(z, f^{n+1}(x)) + ad(f^n(x), f^{n+1}(x)) \\ &\quad + bd(z, f(z)) + cd(f^n(x), z). \end{aligned}$$

Therefore,

$$(1-b)d(z,f(z)) \leq d(z,f^{n+1}(x)) + ad(f^n(x),f^{n+1}(x)) + cd(f^n(x),z).$$

Letting $n \to \infty$, we obtain

$$d(z,f(z)) = 0.$$

Hence,

$$f(z) = z.$$

Therefore, z is a fixed point of f.

We are going to show that the map f has a unique fixed point.

If we assume that $x, y \in X$ with $x \neq y$ are fixed points of the map f, then we have

$$\begin{aligned} d(x,y) &= d(f(x),f(y)) \\ &\leq ad(x,f(x)) + bd(y,f(y)) + cd(x,y) \\ &= cd(x,y). \end{aligned}$$

Since $c < 1$, this implies

$$d(x,y) = 0.$$

Hence,

$$x = y.$$

Therefore, the fixed point is unique.

2.4 Hardy–Rogers Fixed Point Theorem

The Banach contraction principle, while fundamental, requires a strict global contraction condition that is often too rigid in practice. Hardy and Rogers significantly extended this framework by introducing a generalized contraction condition involving five non-negative parameters. Their result encompasses several earlier generalizations, including those of

Kannan, Reich, and Chatterjea as special cases, and thereby provides a unified fixed point theorem for a broad class of self-maps on metric spaces.

Theorem 2.4.1 (**Hardy and Rogers**). *Let (X, d) be a metric space and let*

$$f : X \to X$$

be a map satisfying, for all $x, y \in X$,

$$d(f(x), f(y)) \leq a\, d(x, f(x)) + b\, d(y, f(y)) + c\, d(x, f(y)) + e\, d(y, f(x)) + h\, d(x, y),$$

where

$$a, b, c, e, h \geq 0$$

and

$$\alpha = a + b + c + e + h.$$

Then:

(i) If (X, d) is a complete metric space and $\alpha < 1$, then f has a unique fixed point.

(ii) If (X, d) is compact, f is continuous, and the above condition is replaced by the strict inequality

$$d(f(x), f(y)) < a\, d(x, f(x)) + b\, d(y, f(y)) + c\, d(x, f(y)) + e\, d(y, f(x)) + h\, d(x, y)$$

for all $x \neq y$, and $\alpha = 1$, then f still has a unique fixed point.

Proof. The proof relies on a key auxiliary result under the hypotheses of part (i): there exists a constant

$$\beta < 1$$

such that

$$d(f(x), f^2(x)) \leq \beta\, d(x, f(x)) \qquad \text{for all } x \in X.$$

In particular, one obtains the estimate

$$\beta = \min\left\{\frac{a+c+h}{1-b-c},\ \frac{b+e+h}{1-a-e}\right\}.$$

This is achieved by applying the generalized contraction condition together with the triangle inequality

$$d(f^2(x), x) \geq d(f^2(x), f(x)) - d(f(x), x).$$

Substituting this estimate into the Hardy–Rogers inequality yields an upper bound for

$$d(f^2(x), f(x)),$$

and consequently the above value of β.

Using this estimate repeatedly, for every integer $m > n$ we obtain

$$d(f^m(x), f^n(x)) \leq \frac{\beta^n}{1-\beta}\, d(x, f(x)).$$

Hence the sequence

$$(f^n(x))$$

is a Cauchy sequence. Since (X, d) is complete, there exists

$$z \in X$$

such that

$$z = \lim_{n\to\infty} f^n(x).$$

Applying the Hardy–Rogers condition to the pair

$$(f^n(x), z)$$

and letting $n \to \infty$, we obtain

$$d(z, f(z)) \leq (b + c)\, d(z, f(z)).$$

Since

$$b + c \leq \alpha < 1,$$

it follows that

$$d(z, f(z)) = 0.$$

Therefore,

$$f(z) = z.$$

Hence z is a fixed point of f.

To prove uniqueness, suppose that u and v are fixed points of f. Then

$$f(u) = u, \qquad f(v) = v.$$

Using the Hardy–Rogers condition, we obtain

$$\begin{aligned} d(u, v) &= d(f(u), f(v)) \\ &\leq a\, d(u, f(u)) + b\, d(v, f(v)) \\ &\quad + c\, d(u, f(v)) + e\, d(v, f(u)) + h\, d(u, v) \\ &= (c + e + h)\, d(u, v). \end{aligned}$$

Since

$$c + e + h < \alpha < 1,$$

it follows that

$$d(u, v) = 0.$$

Hence

$$u = v.$$

Therefore the fixed point is unique.

For part (ii), compactness of (X, d) guarantees that the quantity

$$\inf\{d(x, f(x)) : x \in X\}$$

is attained at some point $y \in X$. The strict inequality together with an argument analogous to part (i) forces

$$y = f(y),$$

thereby establishing the existence of a fixed point. Uniqueness follows as before.

Remark 2.4.1. *The Hardy–Rogers condition is sometimes written in a symmetric form by observing that the roles of the coefficients a and b, and also c and e, may be interchanged. This symmetry yields the alternative estimate*

$$\beta = \frac{b + e + h}{1 - a - e},$$

and confirms that the same contraction constant β works for both orientations of the inequality.

Conclusion

This project has undertaken a systematic study of the Banach Contraction Principle and several of its significant generalizations within the framework of complete metric spaces. Beginning with the foundational definitions of metric spaces, Cauchy sequences, contraction mappings, and fixed points, we rigorously established the Banach Fixed Point Theorem and demonstrated its practical utility through applications to systems of linear algebraic equations and the Picard–Lindelöf existence and uniqueness theorem for initial value problems. The second chapter revealed the limitations of the classical Banach contraction condition and motivated the exploration of more flexible fixed point results. The four generalizations examined — the Meir–Keeler contraction, Ćirić's generalized contraction, the Reich theorem, and the Hardy–Rogers theorem — each relax the strict Lipschitz-type condition in a distinct and mathematically meaningful way. The Hardy–Rogers theorem, in particular, subsumes several earlier results as special cases, providing a unified framework for a broad class of self-maps on metric spaces. A recurring theme throughout this study is that completeness of the underlying space and an appropriate contraction-type condition together form the essential ingredients for guaranteeing the existence and uniqueness of a fixed point. These results are not merely of theoretical interest — fixed point theorems underpin existence and uniqueness arguments across nonlinear analysis, differential equations, optimization, and mathematical economics. This study affirms that the Banach Contraction Principle, while foundational, represents only the starting point of a rich and continually evolving theory.

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